```
clear;
clc;
close all;

% DESIGN PARAMETERS
Fs = 10000;
Ap = 0.5;
As = 30;

% ========================================
% 1. LOW-PASS FILTER DESIGN
% ========================================
fp_lp = 1500;
fs_lp = 2000;

wp_lp = fp_lp/(Fs/2);
ws_lp = fs_lp/(Fs/2);

ep2 = 10^(Ap/10)-1;
es2 = 10^(As/10)-1;

% Calculate filter order
N_lp = ceil(acosh(sqrt(es2/ep2))/acosh(ws_lp/wp_lp));

% Calculate passband ripple factor
epsilon = sqrt(ep2);

% Calculate analog prototype poles
mu = asinh(1/epsilon)/N_lp;
k = 1:N_lp;

p_lp = -sinh(mu)*sin((2*k-1)*pi/(2*N_lp)) ...
       + 1i*cosh(mu)*cos((2*k-1)*pi/(2*N_lp));

% Prewarp cutoff frequency
Oc_lp = 2*tan(pi*fp_lp/Fs);

% Scale poles
p_lp = Oc_lp*p_lp;

% Bilinear transformation
z_lp = (2+p_lp)./(2-p_lp);

% Digital filter coefficients
a_lp = real(poly(z_lp));
b_lp = real(poly(-ones(1,N_lp)));

% Normalize DC gain
b_lp = b_lp*(sum(a_lp)/sum(b_lp));

% ========================================
% 2. HIGH-PASS FILTER DESIGN
```

```
% ========================================
fp_hp = 2500;
fs_hp = 1800;

wp_hp = fp_hp/(Fs/2);
ws_hp = fs_hp/(Fs/2);

% Calculate filter order
N_hp = ceil(acosh(sqrt(es2/ep2))/acosh(wp_hp/ws_hp));

% Analog prototype poles
mu_hp = asinh(1/epsilon)/N_hp;
k_hp = 1:N_hp;

p_hp = -sinh(mu_hp)*sin((2*k_hp-1)*pi/(2*N_hp)) ...
       + 1i*cosh(mu_hp)*cos((2*k_hp-1)*pi/(2*N_hp));

% Prewarp passband frequency
Oc_hp = 2*tan(pi*fp_hp/Fs);

% Transform low-pass prototype into high-pass
p_hp = Oc_hp./p_hp;

% Bilinear transformation
z_hp = (2+p_hp)./(2-p_hp);

% Digital filter coefficients
a_hp = real(poly(z_hp));
b_hp = real(poly(ones(1,N_hp)));

% Normalize gain at Nyquist frequency
b_hp = b_hp*(sum(a_hp.*((-1).^(0:N_hp))) / ...
             sum(b_hp.*((-1).^(0:N_hp))));

% ========================================
% DISPLAY NUMERICAL OUTPUTS
% ========================================
fprintf('EXPERIMENT 2: CHEBYSHEV TYPE-I IIR FILTERS\n\n');

fprintf('LOW-PASS FILTER\n');
fprintf('Filter Order: %d\n',N_lp);
fprintf('Passband Edge Frequency: %.2f Hz\n',fp_lp);
fprintf('Numerator Coefficients:\n');
disp(b_lp);
fprintf('Denominator Coefficients:\n');
disp(a_lp);

fprintf('\nHIGH-PASS FILTER\n');
fprintf('Filter Order: %d\n',N_hp);
fprintf('Passband Edge Frequency: %.2f Hz\n',fp_hp);
fprintf('Numerator Coefficients:\n');
disp(b_hp);
fprintf('Denominator Coefficients:\n');
disp(a_hp);
```

```
% ========================================
% FREQUENCY RESPONSE CALCULATION
% ========================================
f = linspace(0,Fs/2,1024);
z = exp(1i*2*pi*f/Fs);

H_lp = polyval(fliplr(b_lp),z.^(-1))./ ...
       polyval(fliplr(a_lp),z.^(-1));

H_hp = polyval(fliplr(b_hp),z.^(-1))./ ...
       polyval(fliplr(a_hp),z.^(-1));

mag_lp = 20*log10(max(abs(H_lp),1e-10));
mag_hp = 20*log10(max(abs(H_hp),1e-10));

phase_lp = angle(H_lp)*180/pi;
phase_hp = angle(H_hp)*180/pi;

% ========================================
% PLOTTING
% ========================================
figure('Name','Chebyshev Type-I IIR Filters');

subplot(2,2,1);
plot(f,mag_lp,'b','LineWidth',1.5);
grid on;
title(['Chebyshev LPF Magnitude (Order N = ',num2str(N_lp),')']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-60 5]);

subplot(2,2,2);
plot(f,phase_lp,'r','LineWidth',1.5);
grid on;
title('Chebyshev LPF Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');

subplot(2,2,3);
plot(f,mag_hp,'b','LineWidth',1.5);
grid on;
title(['Chebyshev HPF Magnitude (Order N = ',num2str(N_hp),')']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-60 5]);

subplot(2,2,4);
plot(f,phase_hp,'r','LineWidth',1.5);
grid on;
title('Chebyshev HPF Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
```

```
EXPERIMENT 2: CHEBYSHEV TYPE-I IIR FILTERS

LOW-PASS FILTER
Filter Order: 7
Passband Edge Frequency: 1500.00 Hz
Numerator Coefficients:
  Columns 1 through 7

    0.0002    0.0011    0.0032    0.0054    0.0054    0.0032    0.0011

  Column 8

    0.0002

Denominator Coefficients:
  Columns 1 through 7

    1.0000   -4.5504    9.9552  -13.2497   11.4770   -6.4420    2.1685

  Column 8

   -0.3390


HIGH-PASS FILTER
Filter Order: 7
Passband Edge Frequency: 2500.00 Hz
Numerator Coefficients:
  Columns 1 through 7

    0.0049   -0.0342    0.1025   -0.1708    0.1708   -0.1025    0.0342

  Column 8

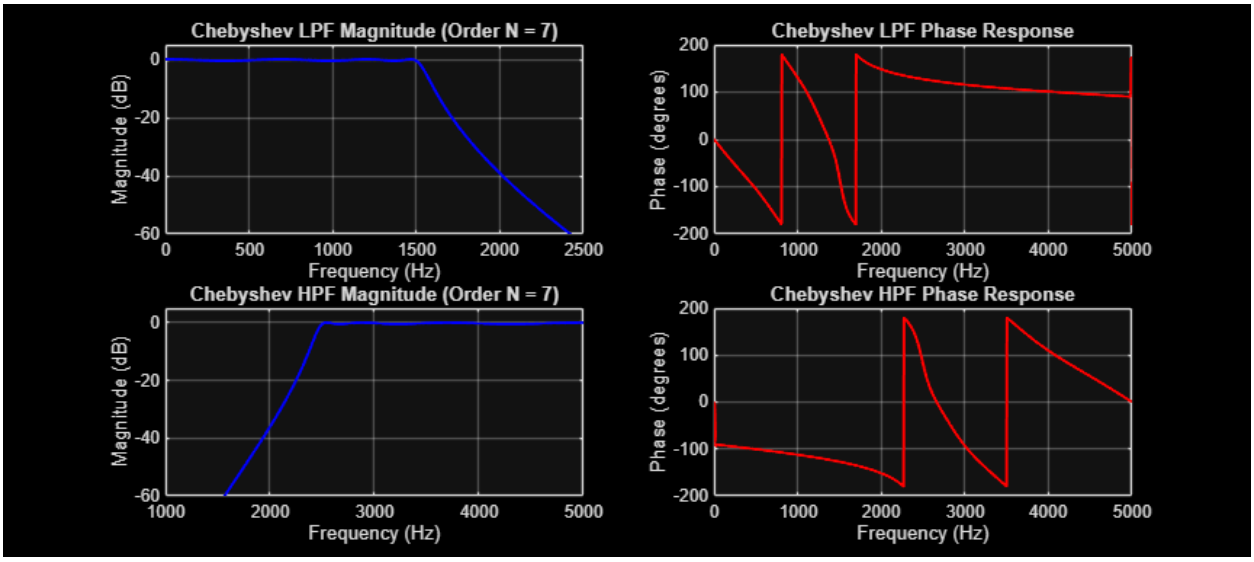
   -0.0049

Denominator Coefficients:
  Columns 1 through 7

    1.0000    1.7706    2.9961    3.1462    2.6310    1.5889    0.6694

  Column 8

    0.1660
```



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